OR-AS Operations Research Applications and Solutions	**Case Name:** Road Drainage Construction with COVID-19 Safety Measures **Project Owner:** Eder Omar Vilca Carrillo	Sector	**Construction**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs
		Risk Analysis	Random simulation One of nine std. scenarios User defined distributions
Processed by	**Zahra Hussaini**		
Date	**2025**	Project Control	Automatic tracking
File Name	**C2025-19**		Tracking based on user input

1. Project description

Project authenticity

This project describes the different activities and their relations in the construction and improvement of the **Alcantarilla Ruta R04** (culvert along the Playa Hermosa - Shama route), including earthworks, concrete structures, drainage systems, and COVID-19 safety measures to ensure proper infrastructure functionality and worker protection. The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	38
Planned Duration (PD)	59 days
Budget At Completion (BAC)	392053.5313 EUR
Renewable Resources	7 Types
Consumable Resources	0 Types

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	70%
Activity Distribution (AD)	65%
Length of Arcs (LA)	0%
Topological Float (TF)	52%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	12.2	14.0	3.3
CRI-rho	12.0	13.9	3.3
CRI-tau	9.1	10.3	2.8

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	13.7	12.4	4.3
CRI-rho	13.7	12.4	4.3
CRI-tau	13.7	12.4	4.3

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	37.6	27.4	0.1
SI	49.7	28.2	-0.5
SSI	4.4	4.9	1.2
CRI-r	11.4	13.7	4.2
CRI-rho	11.6	13.2	3.9
CRI-tau	9.1	10.6	4.2

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	11.7	9.7
PV - SPI	14.3	10.3
PV - SCI	14.7	11.2
ED - 1	10.8	9.0
ED - SPI	13.0	9.0
ED - SCI	13.3	9.6
ES - 1	9.4	8.5
ES - SPI(t)	11.2	9.9
ES - SCI(t)	11.6	10.4

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.6	0.2
CPI	1.1	0.8
SPI	5.0	0.5
SPI(t)	4.6	1.6
SCI	5.6	1.1
SCI(t)	5.3	2.2
0.8 CPI + 0.2 SPI	1.6	1.0
0.8 CPI + 0.2 SPI(t)	1.6	1.1

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 2 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	-639.5	-3.5	1.0	1.0	0.9	1.0
std dev	0	1611.5	6.4	0.0	0.0	0.2	0.0
final	0	500.0	1.0	1.0	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	59 days	Real Duration	60 days	Late	2%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	59.5	0.7	0.4	0.4
PV - SPI	61.5	3.5	1.3	-1.3
PV - SCI	61.5	3.5	1.3	-1.3
ED - 1	59.5	0.7	0.4	0.4
ED - SPI	61.5	3.5	1.3	-1.3
ED - SCI	61.5	3.5	1.3	-1.3
ES - 1	59.5	0.7	0.4	0.4
ES - SPI(t)	69	14.1	7.5	-7.5
ES - SCI(t)	69	14.1	7.5	-7.5

2.3.3.3. Cost forecasting

BAC	329411.9 €	Real Cost	329411.9 €	On budget	0%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	329411.9	0.0	0.0	0.0
CPI	329411.9	0.0	0.0	0.0
SPI	339872.1	14793.0	1.6	-1.6
SPI(t)	380011.2	71558.2	7.7	-7.7
SCI	339872.1	14793.0	1.6	-1.6
SCI(t)	380011.2	71558.2	7.7	-7.7
0.8 CPI + 0.2 SPI	331394.6	2804.0	0.3	-0.3
0.8 CPI + 0.2 SPI(t)	337401.2	11298.7	1.2	-1.2